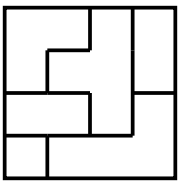
Math Kangaroo Practice - Grades 3-4

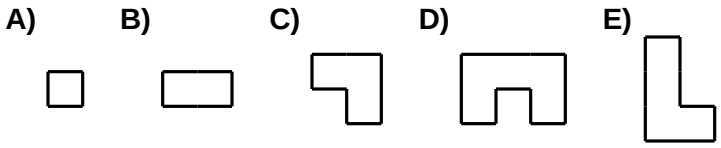
Selected problems from 2002 (#4, #7, #9) and 2003 (#9, #12, #14)
Each original problem is followed by three similar practice problems.

Set 1 - Cutting a square into pieces

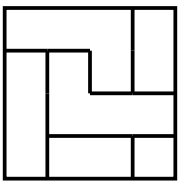
Based on 2002, Problem 4

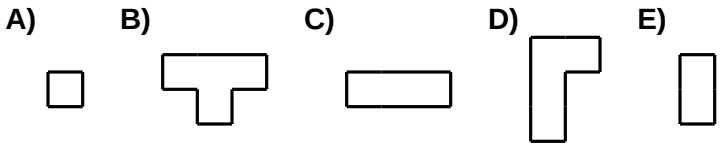
Original 2002 #4. The square was cut along the lines. Which of the shapes below was **not** obtained this way?



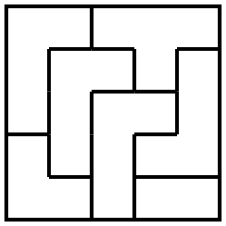


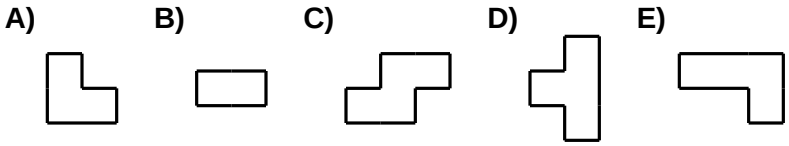
Practice 1.1 The square was cut along the lines. Which of the shapes below was **not** obtained this way?



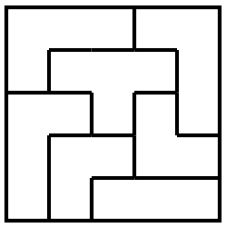


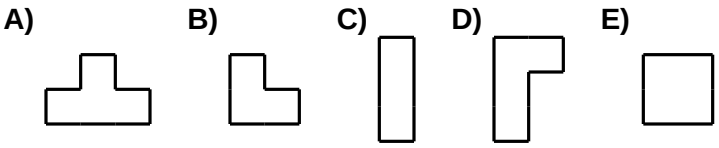
Practice 1.2 This square was cut along the lines. Which shape was **not** obtained?





Practice 1.3 This square was cut along the lines. Which shape was **not** obtained?

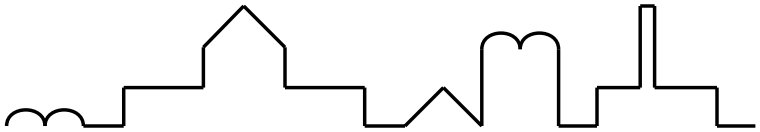


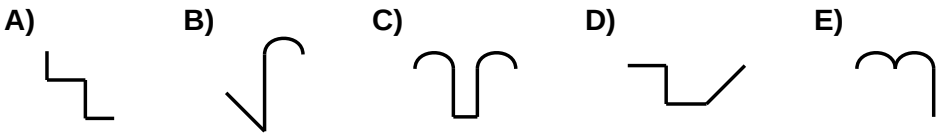


Set 2 - Pieces of a skyline

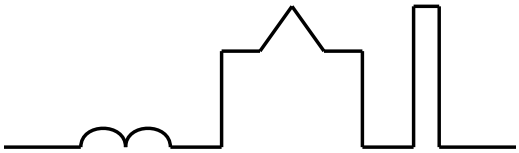
Based on 2002, Problem 7

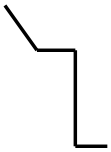
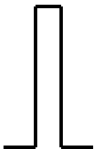
Original 2002 #7. Far away we see the skyline of a castle. Which of the pieces **cannot** belong to the skyline?



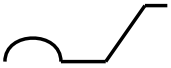
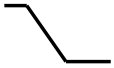


Practice 2.1 Here is the skyline of a town. Which piece **cannot** belong to it?

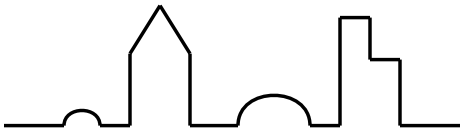


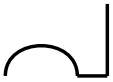
A)
B) 
C)
D)
E) 

Practice 2.2 Here is the skyline of a castle. Which piece **cannot** belong to it?

A)
B) 
C) 
D)
E)

Practice 2.3 Here is a skyline with a small arch, a tower and a dome. Which piece **cannot** belong to it?

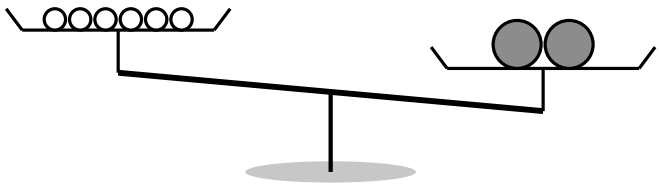


A) 
B)
C)
D)
E)

Set 3 - Balance scales

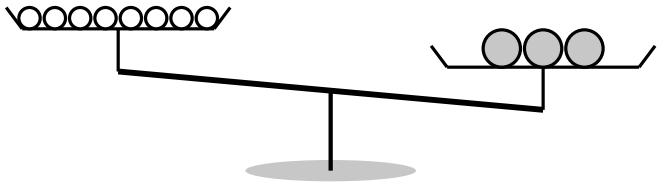
Based on 2002, Problem 9

Original 2002 #9. On one plate of a balance there are 6 oranges (small circles) and on the other there are melons (big circles). When we put one more melon, exactly like the others, on the orange plate, the balance is level. The weight of one melon is the same as:



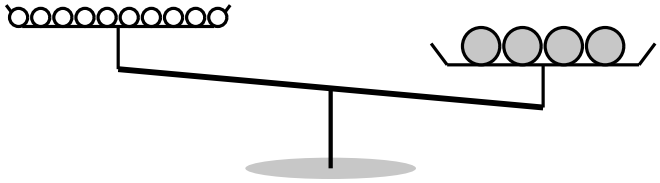
A) 2 oranges **B)** 3 oranges **C)** 4 oranges
D) 5 oranges **E)** 6 oranges

Practice 3.1 On one plate there are 8 plums (small circles) and on the other 3 lemons (gray circles). When we put one more lemon on the plum plate, the balance is level. One lemon weighs the same as:



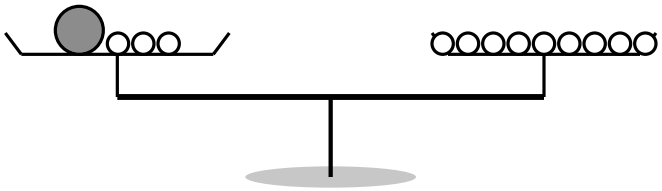
A) 2 plums **B)** 3 plums **C)** 4 plums
D) 5 plums **E)** 8 plums

Practice 3.2 On one plate there are 10 cherries (small circles) and on the other 4 apples (gray circles). When we put 2 more apples on the cherry plate, the balance is level. One apple weighs the same as:



A) 2 cherries **B)** 3 cherries **C)** 4 cherries
D) 5 cherries **E)** 10 cherries

Practice 3.3 The balance is level. On the left plate there is 1 melon (big circle) and 3 oranges; on the right plate there are 9 oranges. How many oranges weigh the same as **2 melons**?

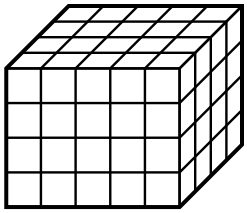


A) 6 **B)** 8 **C)** 9 **D)** 12 **E)** 18

Set 4 - Red outside, blue inside

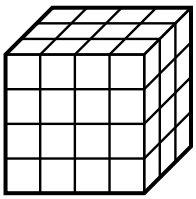
Based on 2003, Problem 9

Original 2003 #9. Chris built the brick in the picture using red and blue cubes of the same size. The outside of the brick is completely red, but all cubes used inside are blue. How many blue cubes did Chris use?



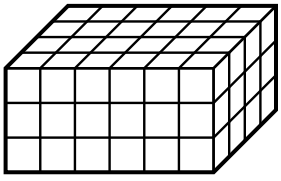
A) 12 **B)** 24 **C)** 36 **D)** 40 **E)** 48

Practice 4.1 Maya built this brick (4 cubes wide, 4 cubes tall, 3 cubes deep). The outside is completely red and all the cubes inside are blue. How many blue cubes did she use?



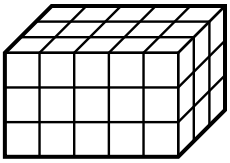
A) 2 **B)** 4 **C)** 6 **D)** 8 **E)** 12

Practice 4.2 Leo built this brick (6 cubes wide, 3 cubes tall, 4 cubes deep). The outside is completely red and all the cubes inside are blue. How many blue cubes did he use?



A) 4 **B)** 6 **C)** 8 **D)** 12 **E)** 16

Practice 4.3 Anna built this brick (5 cubes wide, 3 cubes tall, 3 cubes deep). The outside is completely red and the cubes inside are blue. How many **red** cubes did she use?

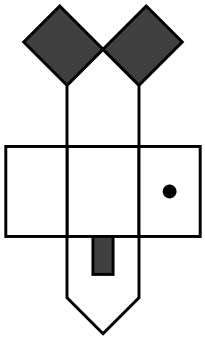


A) 30 **B)** 36 **C)** 39 **D)** 42 **E)** 45

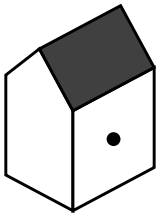
Set 5 - Folding a house

Based on 2003, Problem 12

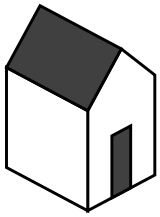
Original 2003 #12. The picture has been drawn on paper and cut out to make a house. Which of the houses did it become? *(The dark parts at the top of the picture are the roof. The shape in the middle is the floor.)*



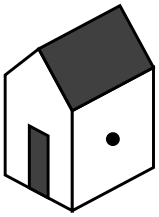
A)



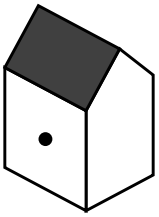
B)



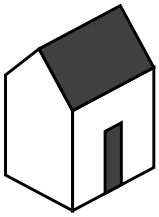
C)



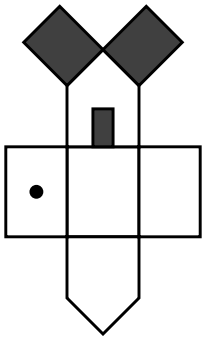
D)

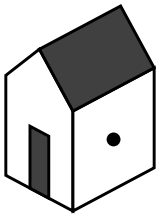


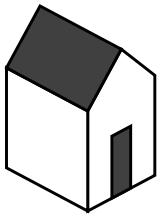
E)

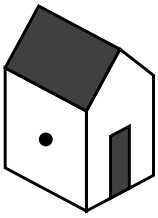


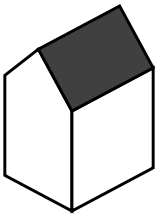
Practice 5.1 This picture is cut out and folded to make a house. Which house does it become? *(The dark parts at the top of the picture are the roof. The shape in the middle is the floor.)*

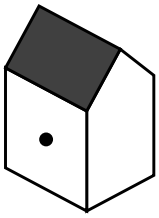


A) 

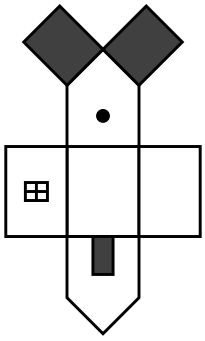
B) 

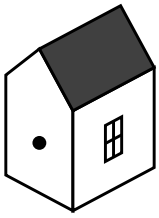
C) 

D) 

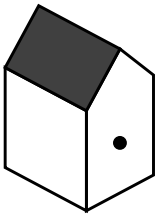
E) 

Practice 5.2 This picture is cut out and folded to make a house. Which house does it become? *(The dark parts at the top of the picture are the roof. The shape in the middle is the floor.)*

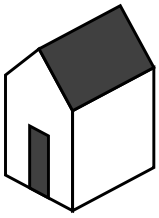


A) 

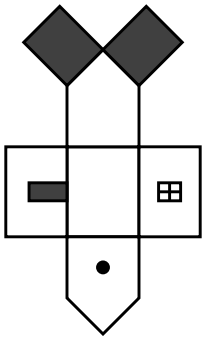
B)

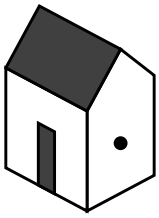
C) 

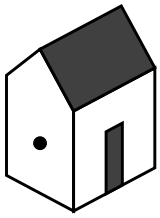
D)

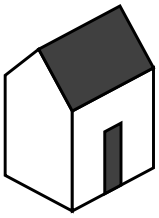
E) 

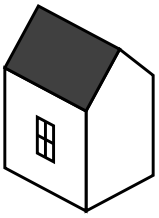
Practice 5.3 This picture is cut out and folded to make a house. Which house does it become? *(The dark parts at the top of the picture are the roof. The shape in the middle is the floor.)*

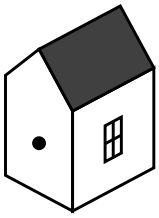


A) 

B) 

C) 

D) 

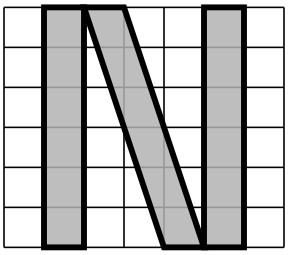
E) 

Set 6 - Area of a letter on squared paper

Based on 2003, Problem 14

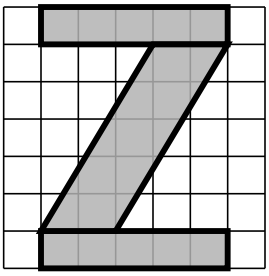
Original 2003 #14. If the side of each little square is 1 cm, what is the area of the letter N (in square cm)?

A) 14 **B)** 15 **C)** 16 **D)** 17 **E)** 18



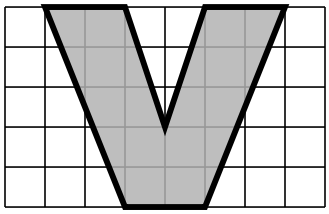
Practice 6.1 If the side of each little square is 1 cm, what is the area of the letter Z (in square cm)?

A) 16 **B)** 18 **C)** 20 **D)** 22 **E)** 24

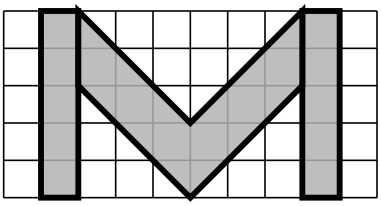


Practice 6.2 If the side of each little square is 1 cm, what is the area of the letter V (in square cm)?

A) 15 **B)** 16 **C)** 17 **D)** 18 **E)** 20



Practice 6.3 If the side of each little square is 1 cm, what is the area of the letter M (in square cm)?



A) 18 **B)** 20 **C)** 22 **D)** 24 **E)** 26

Answer Key & Explanations

For practice_2002_2003.pdf (Selected problems from 2002 (#4, #7, #9) and 2003 (#9, #12, #14))

Solutions & kid-friendly explanations

2002 #4: D The pieces are: 1 small square, 2 dominoes, an L of 3, a T of 4 and an L of 4. There is no U-shaped piece.

1.1: B Pieces: two L-shapes of 4, a straight 3, two dominoes, one small square. No T-shape.

1.2: C Pieces: four L-shapes of 4, one T, one L of 3, one domino. There is no zig-zag (S) piece.

1.3: E Pieces: three L-shapes of 4, one T, two L-shapes of 3, one straight 3. No 2x2 square.

2002 #7: C Wherever there are two arches, they sit right next to each other. There is never a U-shaped dip between two arches.

2.1: A In the skyline there is a flat part between the wall and the pointed roof. Piece A has the roof sitting straight on the wall.

2.2: E The castle wall has TWO notches at the top. Piece E shows a whole tower top with only one notch.

2.3: D After each arch the line goes flat and then UP. Piece D goes down after the arch.

2002 #9: E 6 oranges + 1 melon = 2 melons. Take one melon off each side: 6 oranges = 1 melon.

3.1: C 8 plums + 1 lemon = 3 lemons, so 8 plums = 2 lemons, so 1 lemon = 4 plums.

3.2: D 10 cherries + 2 apples = 4 apples, so 10 cherries = 2 apples, so 1 apple = 5 cherries.

3.3: D Take 3 oranges off each side: 1 melon = 6 oranges, so 2 melons = 12 oranges.

2003 #9: A Brick is 5 x 4 x 4. Remove the outer layer: inside is 3 x 2 x 2 = 12 blue cubes.

4.1: B Inside is (4-2) x (4-2) x (3-2) = 2 x 2 x 1 = 4.

4.2: C Inside is (6-2) x (3-2) x (4-2) = 4 x 1 x 2 = 8.

4.3: D Total = 5 x 3 x 3 = 45. Inside (blue) = 3 x 1 x 1 = 3. Red = 45 - 3 = 42.

2003 #12: A Stand in front of the door wall: the dot wall is on your LEFT. So from the back (no door), the dot is on the right of the plain end wall - house A.

5.1: C The door is on the end wall that has the roof attached. Facing that door, the dot wall is on your left.

5.2: D Facing the door wall, the window wall is on your right and the plain wall on your left. Only D shows that.

5.3: B Facing the dot end wall: the door wall is on your right, the window wall on your left. B shows the dot wall with the door wall to its right.

2003 #14: E Each side bar is 6. The slanted part is a parallelogram 1 wide and 6 tall = 6 (cut-off triangles fit together). 6+6+6 = 18.

6.1: C Top bar 5 + bottom bar 5 + slanted part (2 wide, 5 tall) 10 = 20.

6.2: C Without the notch the shape is a 2 x 5 rectangle (10) plus two side triangles (each half of 2 x 5 = 5), so 20. The white triangle notch at the top is half of 2 x 3 = 3. 20 - 3 = 17.

6.3: C Two side bars: 5 + 5 = 10. The middle V is two slanted strips, each 3 across and 2 tall = 6. 10 + 6 + 6 = 22.